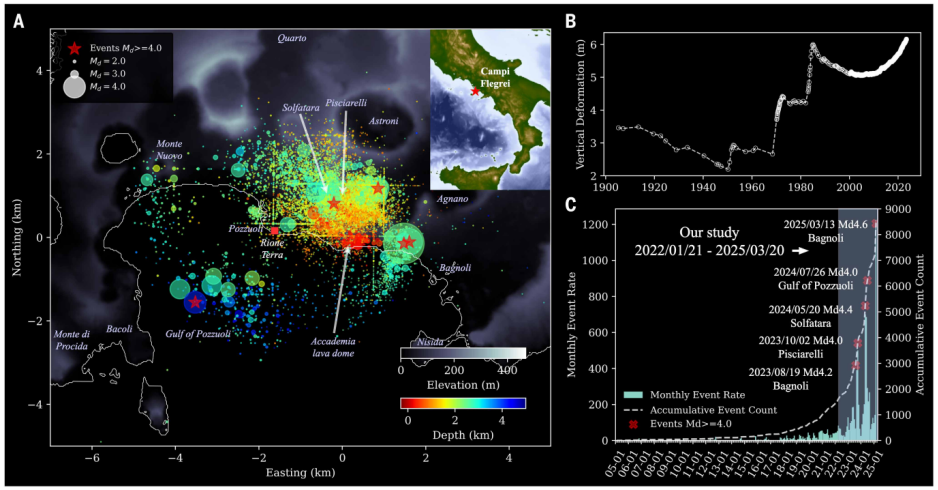
Figure 1: Seismicity and ground displacement

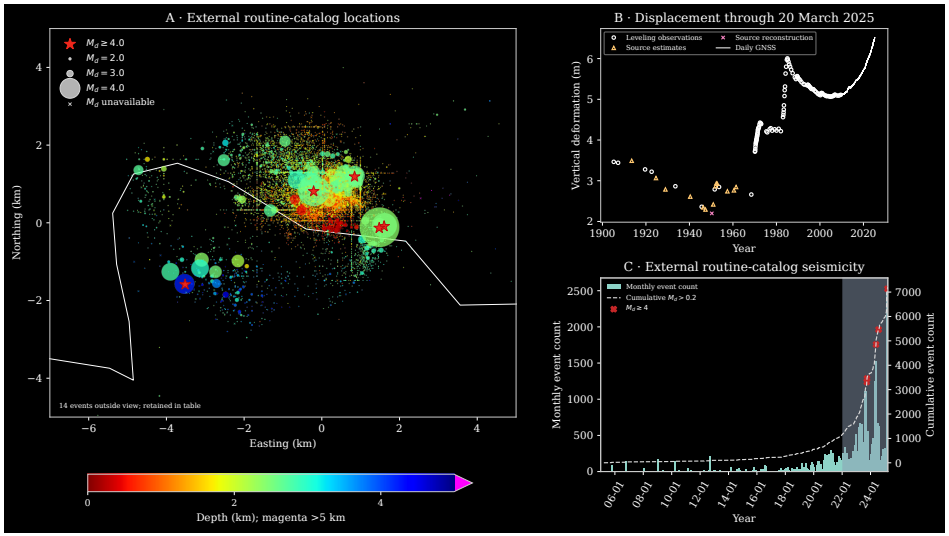
Published reference



Tan et al. (2025), tanmodel.pdf, p. 2.
DOI: 10.1126/science.adw9038.

Independent result

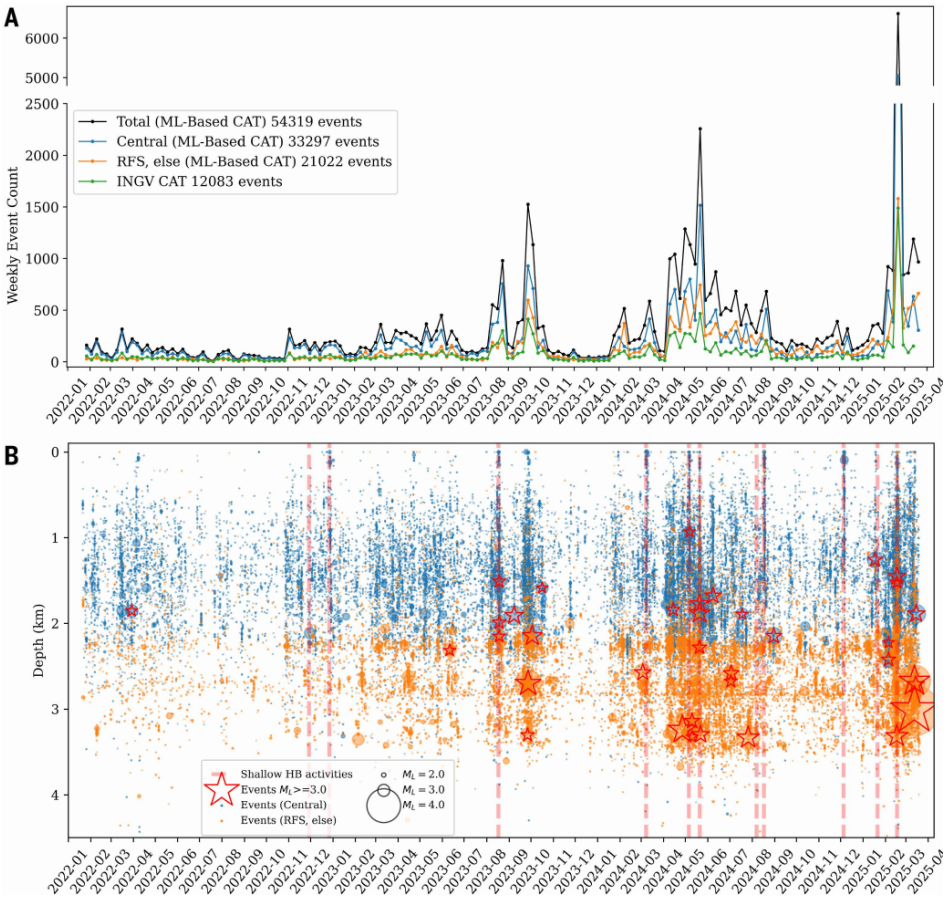
Panel 1A–C — partially reproduced



External routine-catalog context and source-labeled leveling/GNSS observations. These earthquakes are not independent detections. Historical snapshot and datum differences remain.

Figure 2: Temporal evolution of seismicity

Published reference

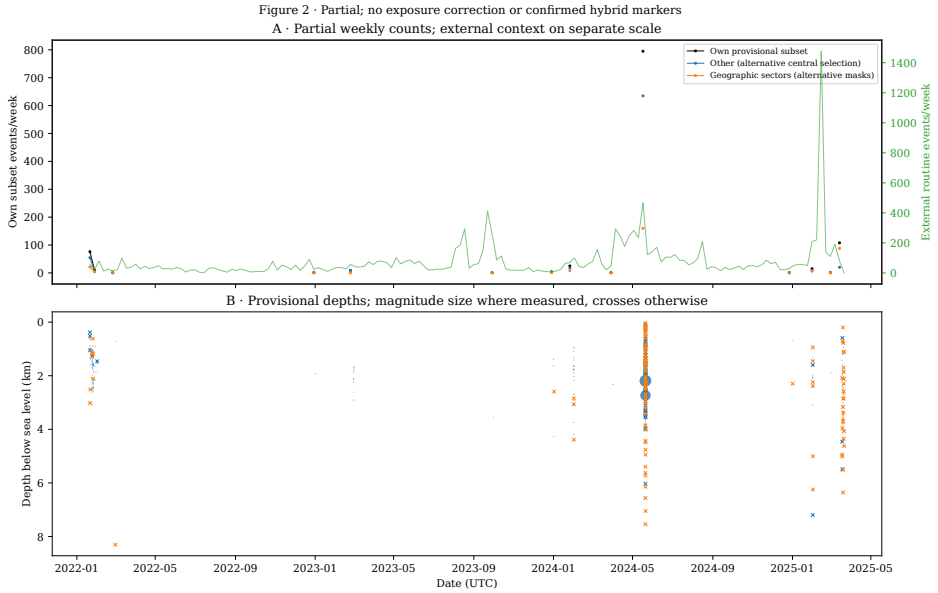


Tan et al. (2025), tanmodel.pdf, p. 3.

DOI: 10.1126/science.adw9038.

Independent result

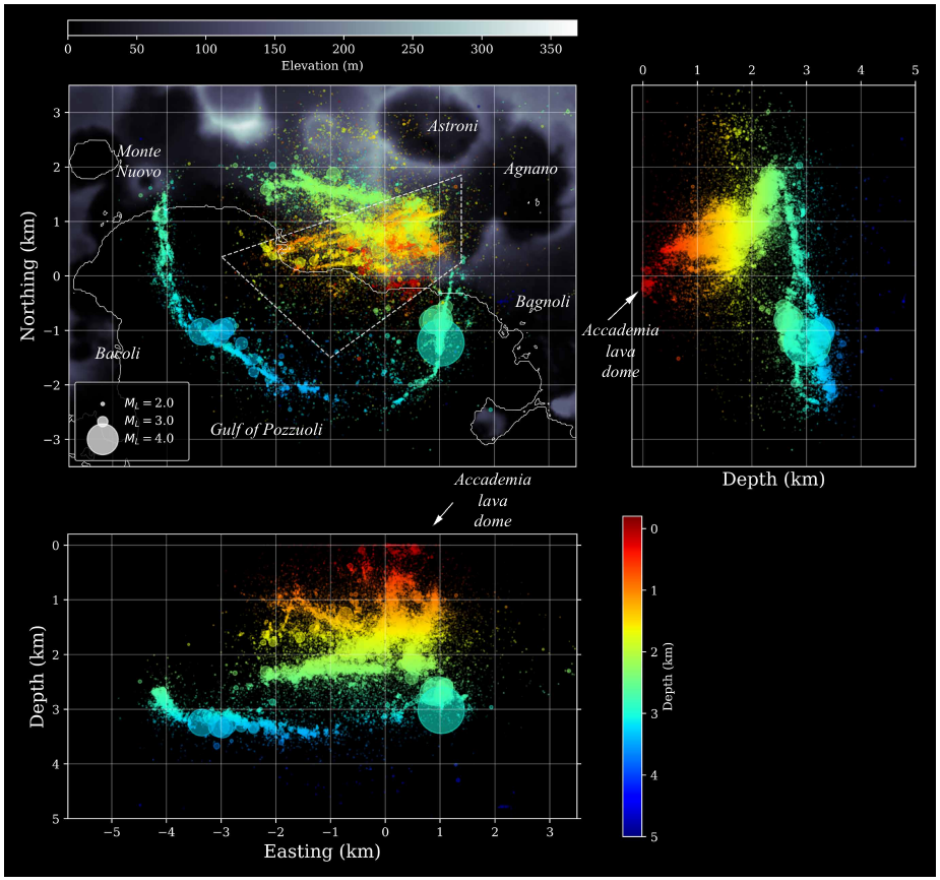
Panel 2; current provisional independent calculation — partially reproduced



1050 own relocated trial events in weekly counts and depth versus time. Unobserved weeks are not measured zero rates. Explicit alternative geographic sectors; routine counts appear only as external context.

Figure 3: Spatial distribution of seismicity

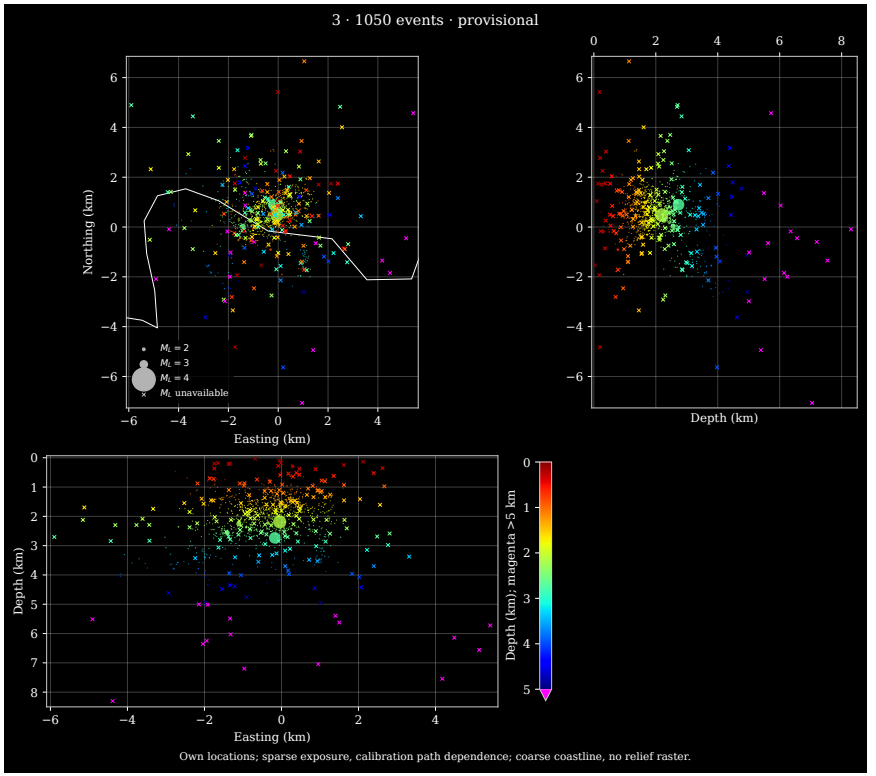
Published reference



Tan et al. (2025), tanmodel.pdf, p. 4.
DOI: 10.1126/science.adw9038.

Independent result

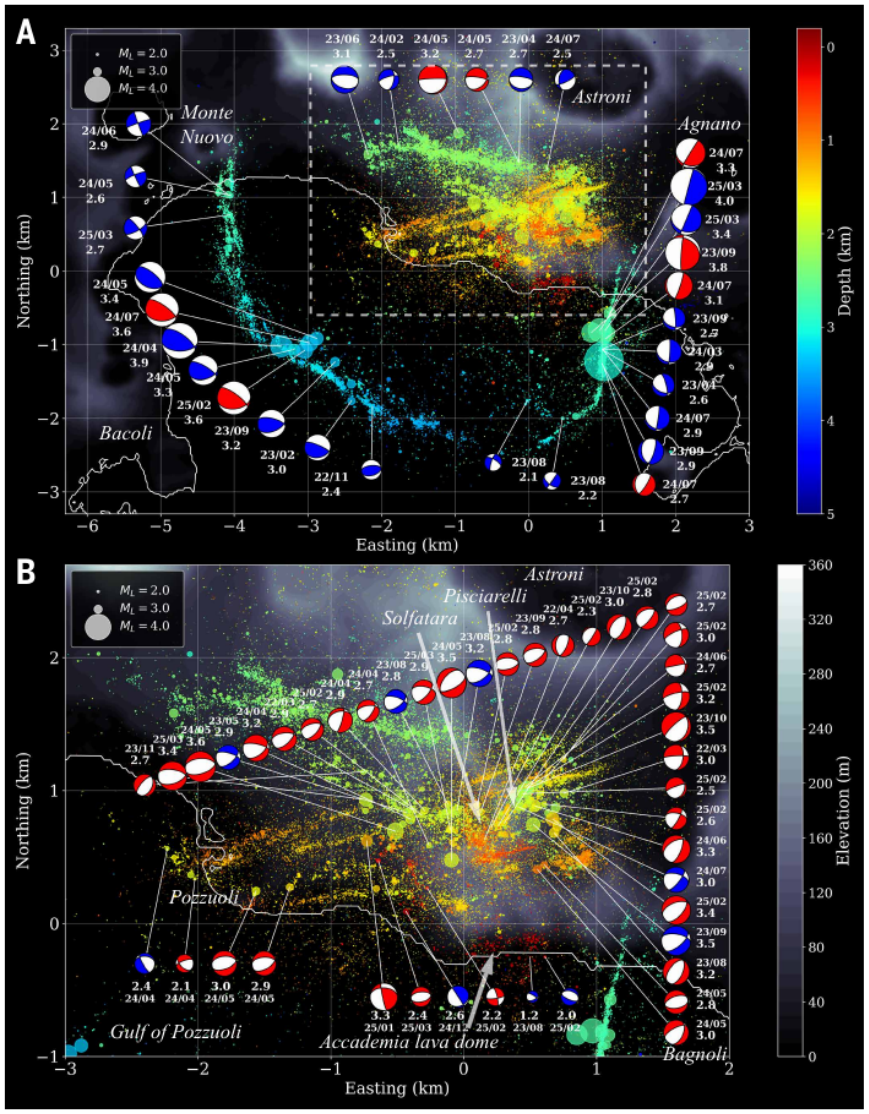
Panel 3; current provisional independent calculation — partially reproduced



1050 own relocated trial events, 0.031–8.303 km below sea level. 16 points exceed 5 km and retain magenta over-range colors. Own provisional magnitude sets marker area when available; crosses mark missing magnitudes. Ring geometry and depth limits remain unvalidated.

Figure 4: Focal mechanisms of larger events

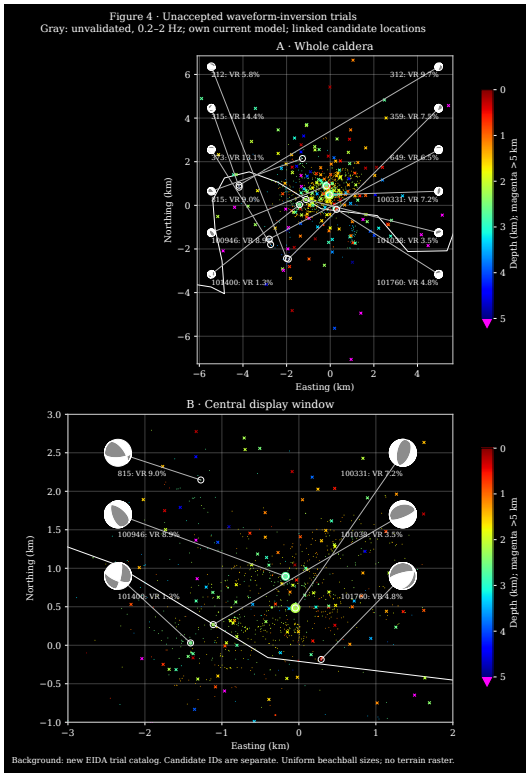
Published reference



Tan et al. (2025), tanmodel.pdf, p. 5.
DOI: 10.1126/science.adw9038.

Independent result

Panel 4; current provisional independent calculation — partially reproduced



12 unaccepted own FK/gCAP trial fits using newly available EIDA records and the current own model and explicitly linked own candidate geometry. 99 AI-reviewed polarities, 108 AI-reviewed ratios; no human review. SKHASH returns 2 solutions. Low-band variance reduction 1.3–14.4%; orientations change 45.0–95.0 degrees between frequency bands. Gray beachballs denote unvalidated trials; background is the new independent catalog, never joined to candidates by integer ID. SKHASH returns 2 quality-D trials. No scientifically accepted mechanism.

Figure S1: Recorded station availability

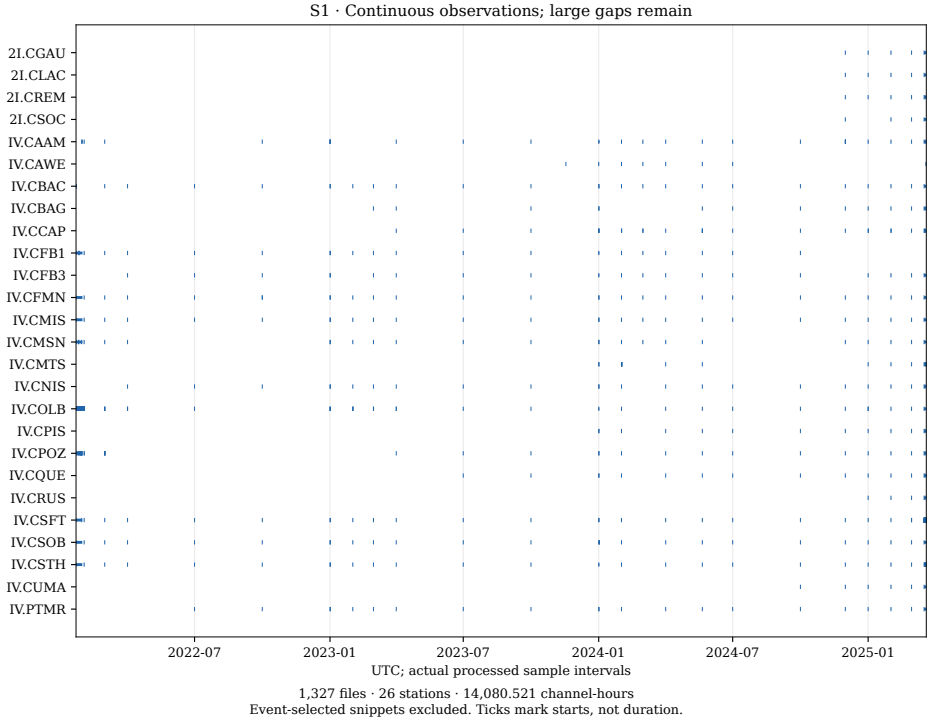
Published reference



Tan et al. (2025), tanmodelsupp.pdf, p. 12.
DOI: 10.1126/science.adw9038.

Independent result

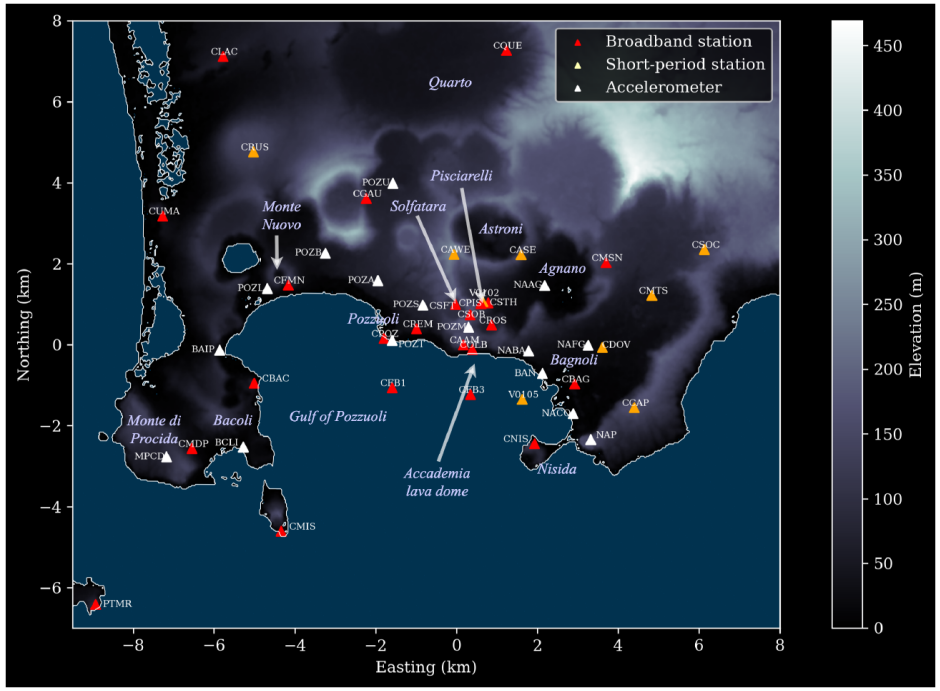
Panel S1; current provisional independent calculation — partially reproduced



1,327 eligible continuous files, 14,080.521 processed channel-hours, 26 stations. Actual merged intervals only; ticks identify starts, not duration. Failed and pending requests and event-selected snippets are excluded. Large temporal/network gaps remain.

Figure S2: Station locations

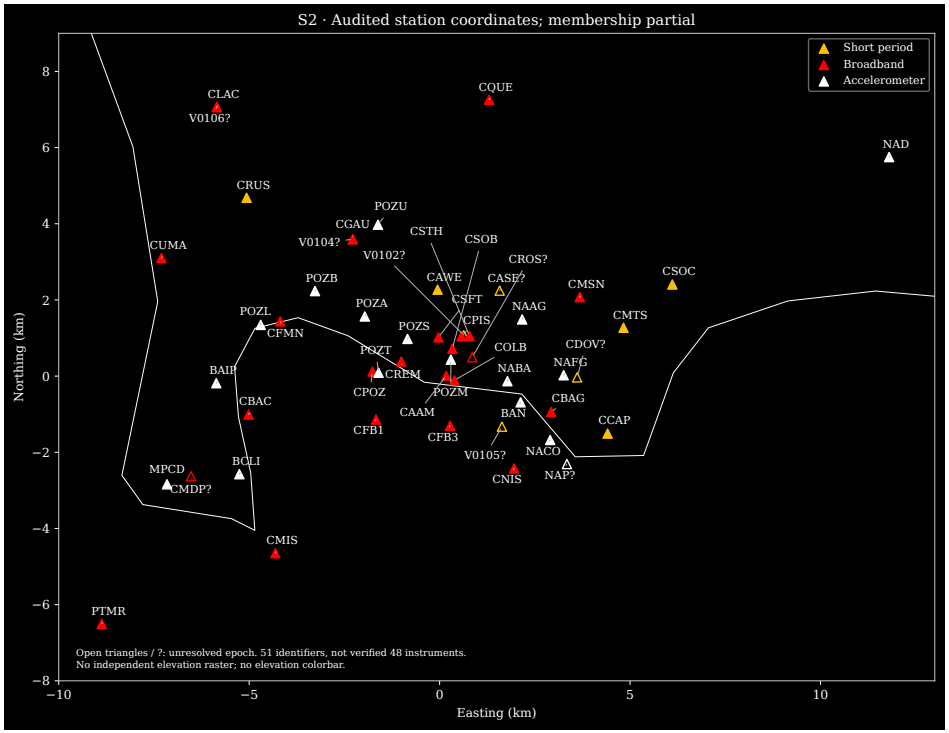
Published reference



Tan et al. (2025), tanmodelsupp.pdf, p. 13.
DOI: 10.1126/science.adw9038.

Independent result

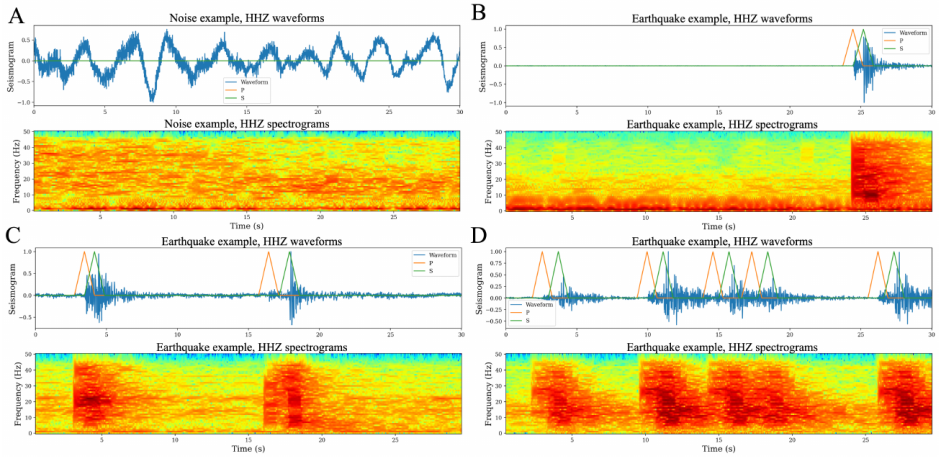
Panel S2 — partially reproduced



Partial station inventory: 51 released identifiers audited against official study-period response epochs. Filled triangles: 42 official coordinate matches; open triangles: 9 released coordinates with unresolved epochs or channel codes. Two coincident-coordinate pairs remain distinct identifiers. No verified geographic outlier. Local WGS84 projection and clipped coastline; the original 48-instrument membership is not established. Colors distinguish broadband (red), short period (yellow) and accelerometer (white) instruments; no independent relief raster is available.

Figure S3: Training examples

Published reference



Tan et al. (2025), tanmodelsupp.pdf, p. 14.
DOI: 10.1126/science.adw9038.

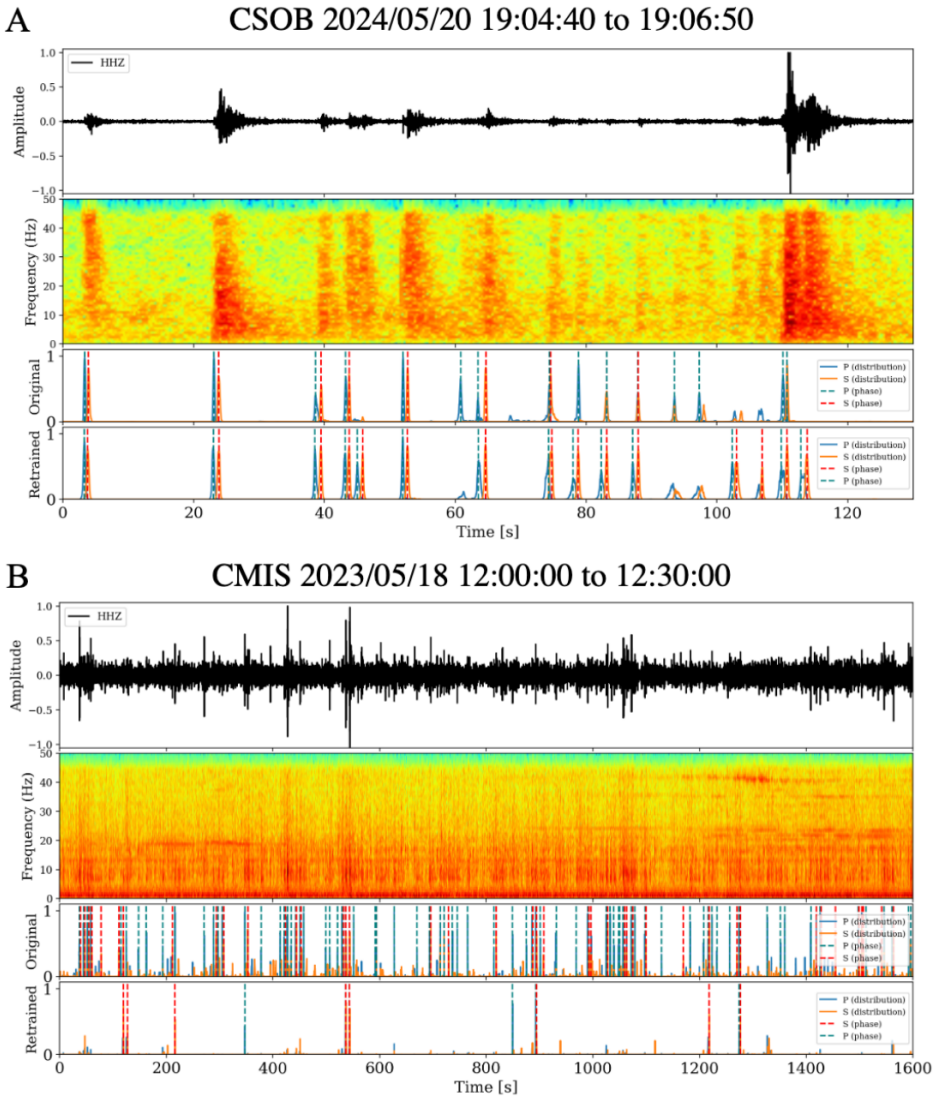
Independent result

Missing: S3A, S3B, S3C, S3D — not reproduced

Panels A–D require the specified training traces, annotations and augmentation examples. These were not reconstructed; downstream use of trained weights is not retraining.

Figure S4: PhaseNet benchmark examples

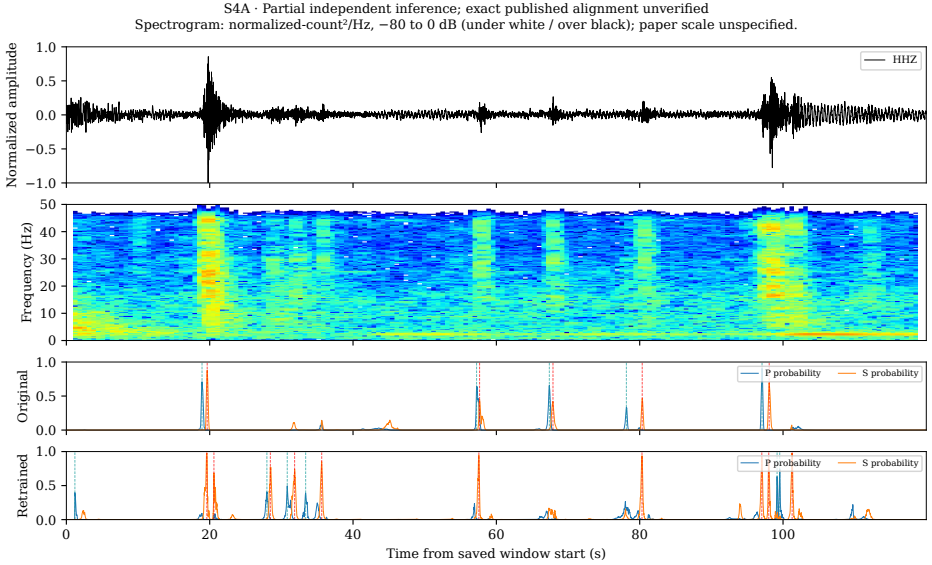
Published reference



Tan et al. (2025), tanmodelsupp.pdf, p. 15.
DOI: 10.1126/science.adw9038.

Independent result

Panel S4A — partially reproduced



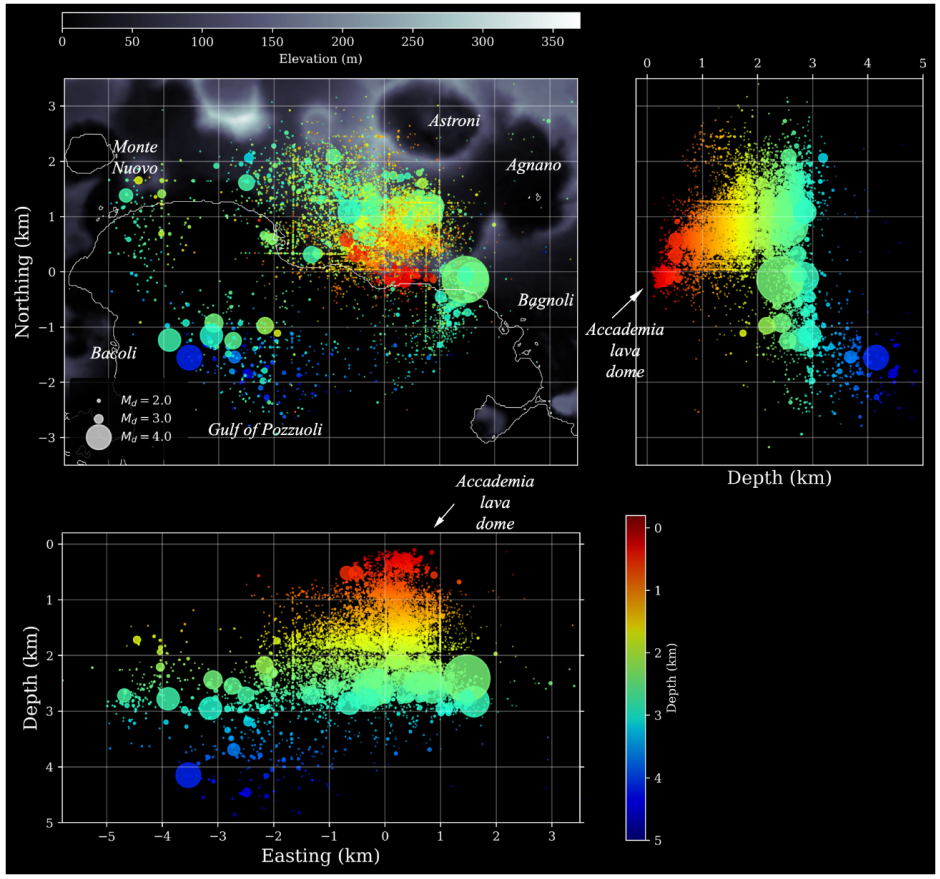
Partial S4A inference comparison on original CSOB observations, 20 May 2024 20:10:03–20:12:03 UTC. The caption's two-hour offset is interpreted literally; exact published alignment is unverified. Original weights: 5 P and 5 S; retrained weights: 6 P and 10 S. No labeled accuracy or retraining claim.

Missing: S4B — not reproduced

S4B: exact CMIS noise-window timing and reference labels are unavailable. Other calendar-window diagnostics are not inserted as that panel.

Figure S5: Routine-catalog locations

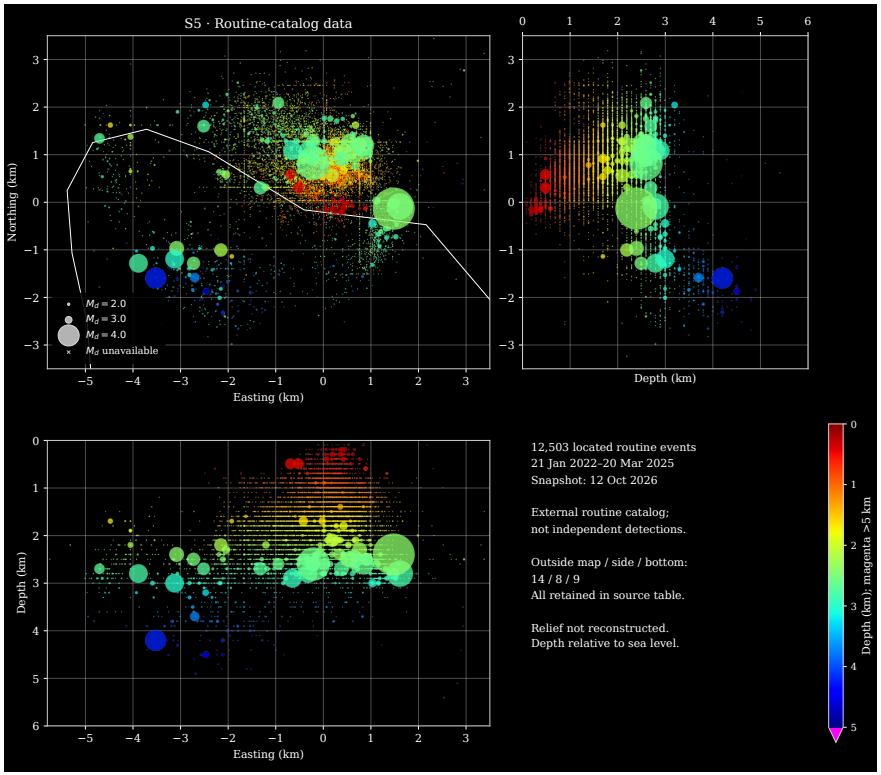
Published reference



Tan et al. (2025), tanmodelsupp.pdf, p. 16.
DOI: 10.1126/science.adw9038.

Independent result

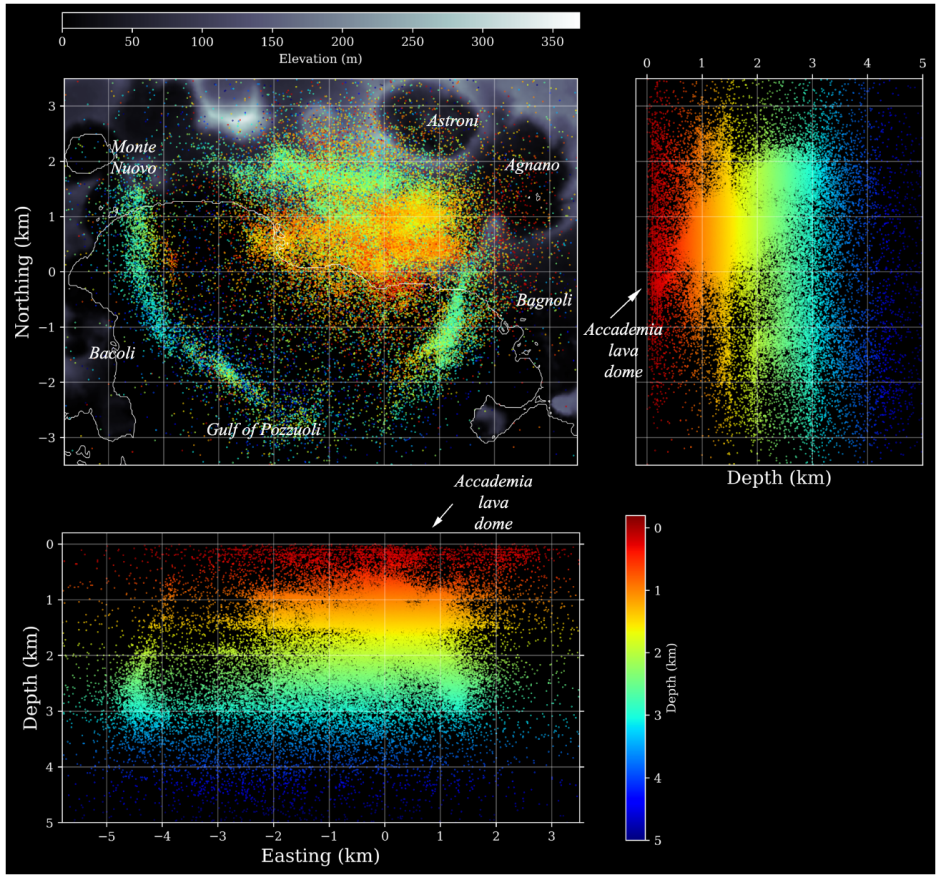
Panel S5, all views — partially reproduced



External routine-catalog data, independently selected and plotted: 12,503 located events, 21 January 2022–20 March 2025. This is 420 more than the published historical snapshot. No author-derived catalog is used to adjust that difference; these are not independent detections. Depth colors: fixed 0–5 km, red to blue; magenta >5 km, no data clipping. S5 limits/ticks are printed; 1A endpoints are unlabeled and 0–5 km is an explicit approximation. Circle area law is an explicit display assumption, not recovered exactly. Missing magnitudes are crosses. Routine events are external context.

Figure S6: Initial VELEST locations

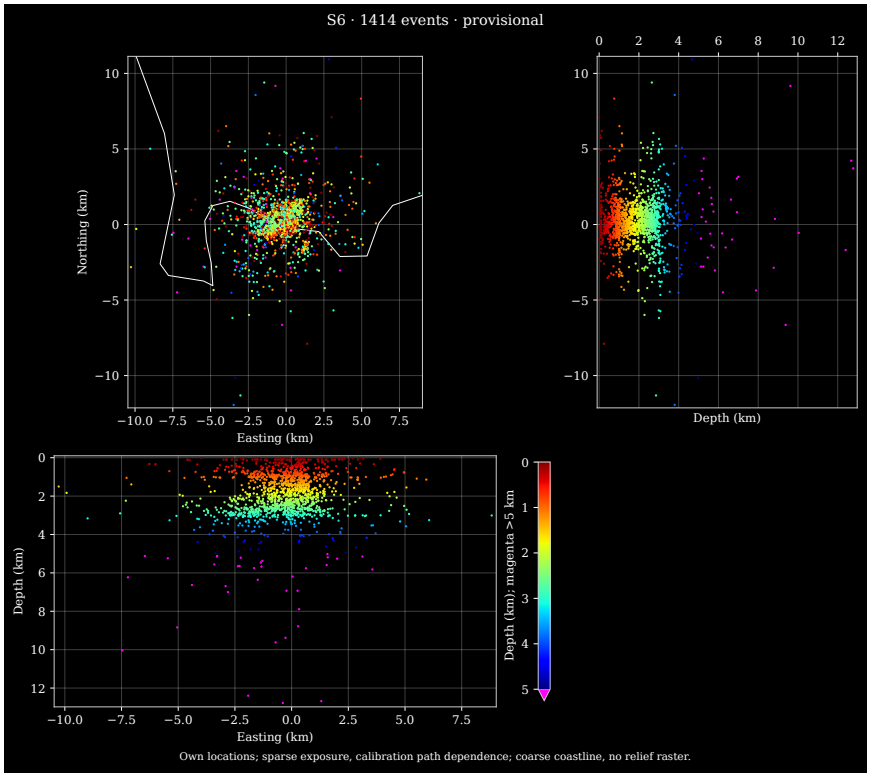
Published reference



Tan et al. (2025), tanmodelsupp.pdf, p. 17.
DOI: 10.1126/science.adw9038.

Independent result

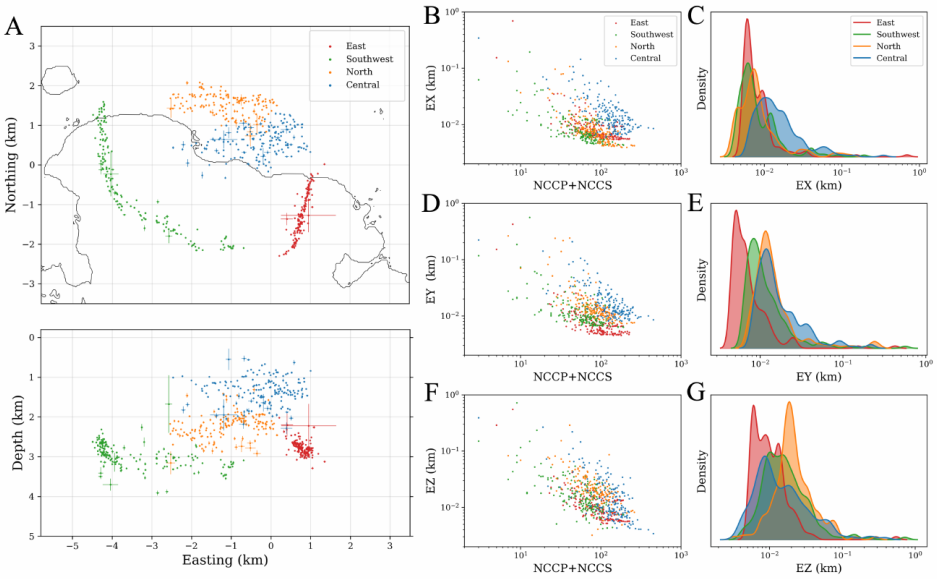
Panel S6; current provisional independent calculation — partially reproduced



1,414 own five-pass locations pass numerical filters. Independent alternative-initialized profile; failed calendar holdout, provisional only. This calculation is separate from the prior grid-initialized TomoDD checkpoint. Depth color limits 0–5 km, integer ticks, jet_r, magenta over-range; no depth clipping. Uniform point area is a display choice.

Figure S7: Location uncertainties

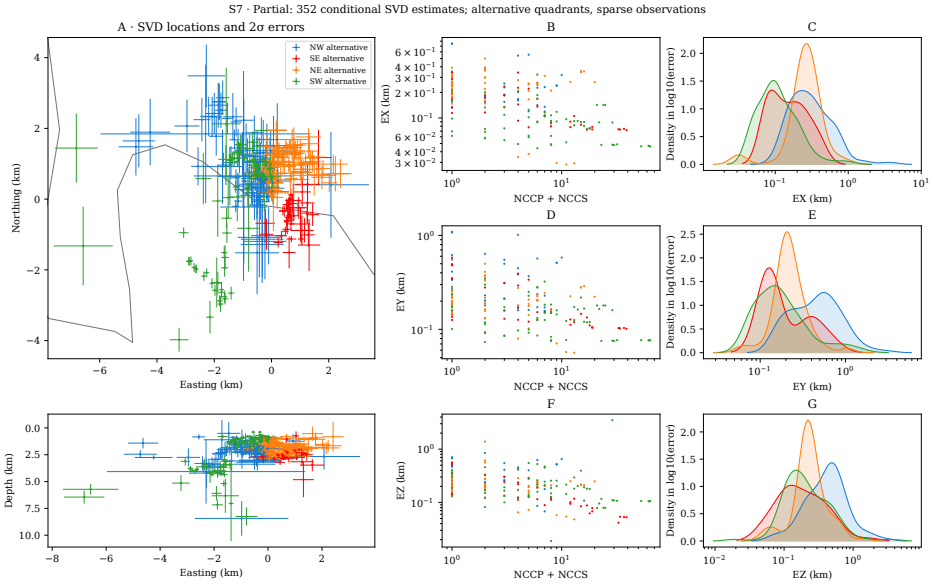
Published reference



Tan et al. (2025), tanmodelsupp.pdf, p. 18.
DOI: 10.1126/science.adw9038.

Independent result

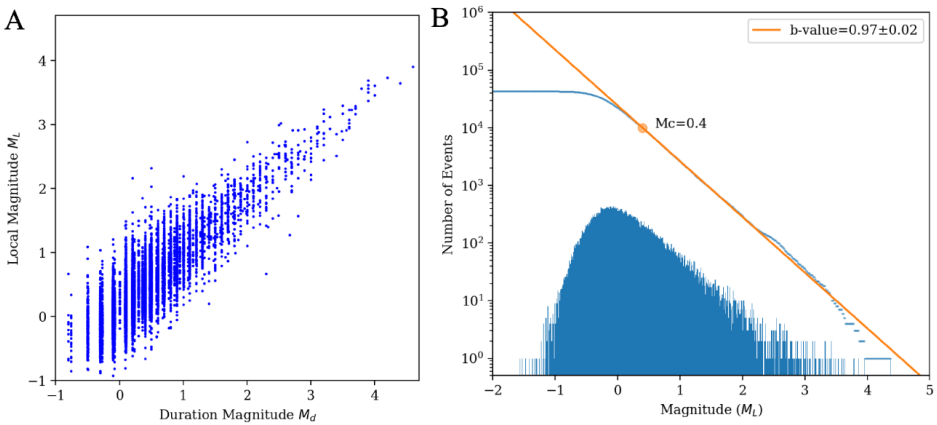
Panel S7; current provisional independent calculation — partially reproduced



352 conditional HypoDD SVD estimates in four explicitly alternative geographic subsets. Maximum vertical 2-sigma error 3.499 km. Regional samples are capped at the written 174/165/175/166 events with disclosed seed 42; membership is saved. Model and phase-identification biases are not included; zero correlation counts omitted only from logarithmic axes.

Figure S8: Magnitude calibration and frequency distribution

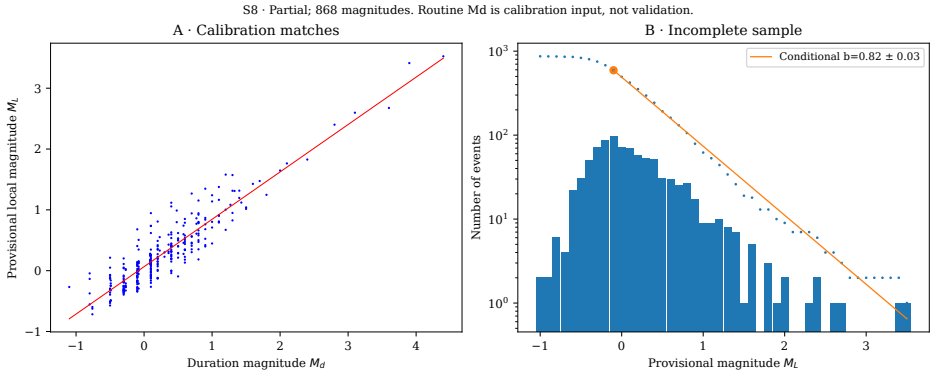
Published reference



Tan et al. (2025), tanmodelsupp.pdf, p. 19.
DOI: 10.1126/science.adw9038.

Independent result

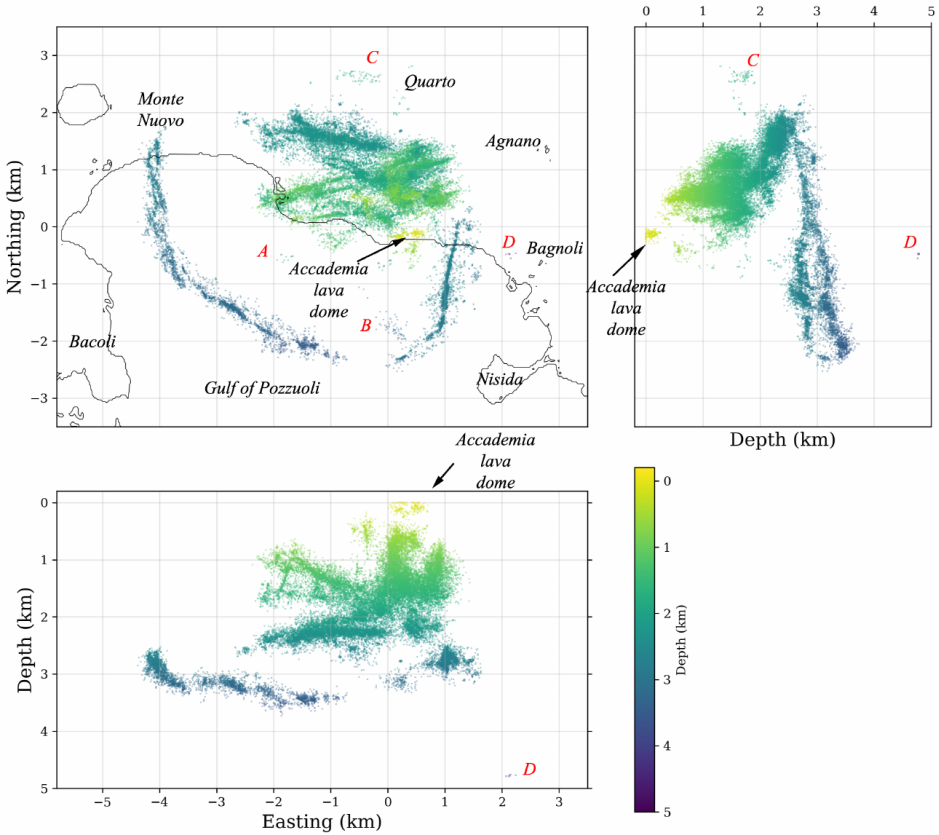
Panel S8; current provisional independent calculation — partially reproduced



868 own provisional magnitudes from 167234 amplitude equations and 374 external duration-magnitude priors; 308 measured ML–Md pairs are displayed. Conditional Mc=-0.10, b=0.822 +/- 0.032, N=592. External Md is calibration, not validation. Discontinuous sampling prevents a population-completeness claim.

Figure S9: Best-located earthquake subset

Published reference



Tan et al. (2025), tanmodelsupp.pdf, p. 20.
DOI: 10.1126/science.adw9038.

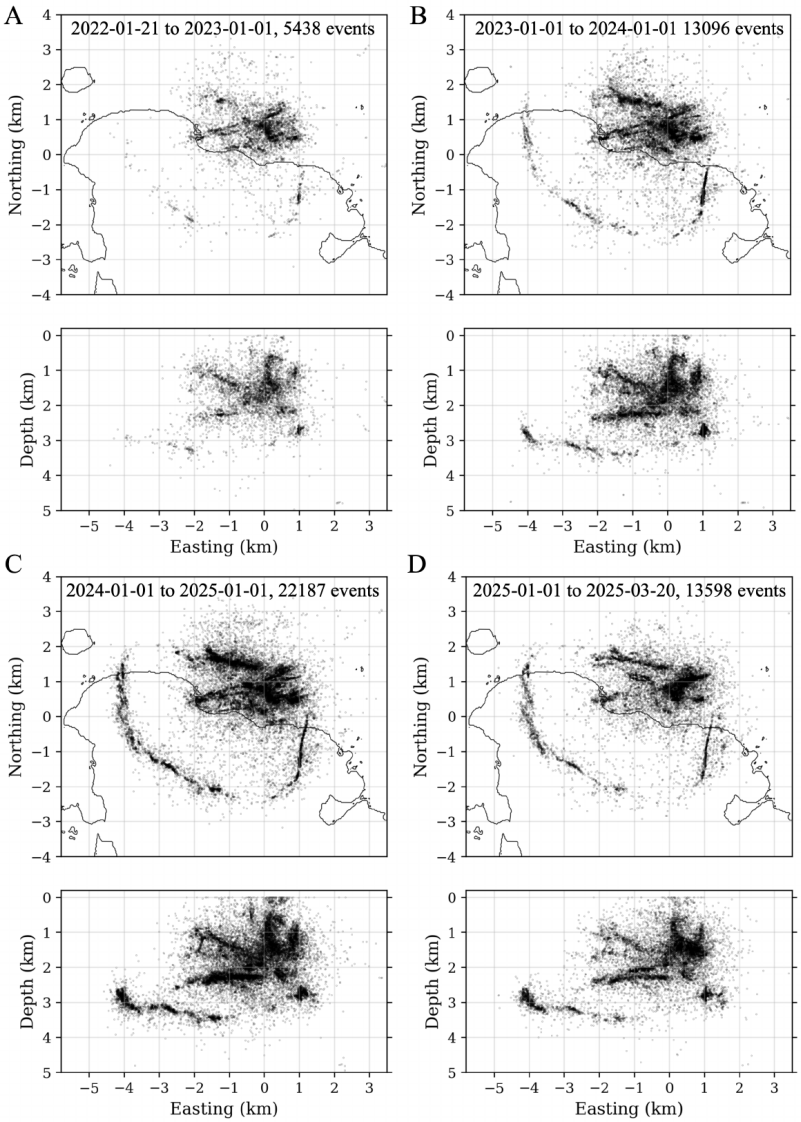
Independent result

Missing: S9/map, S9/northing-depth, S9/easting-depth — not reproduced

No own retained event satisfies NCCP+NCCS >=100; no coordinates can be plotted for this selection.

Figure S10: Seismicity by year

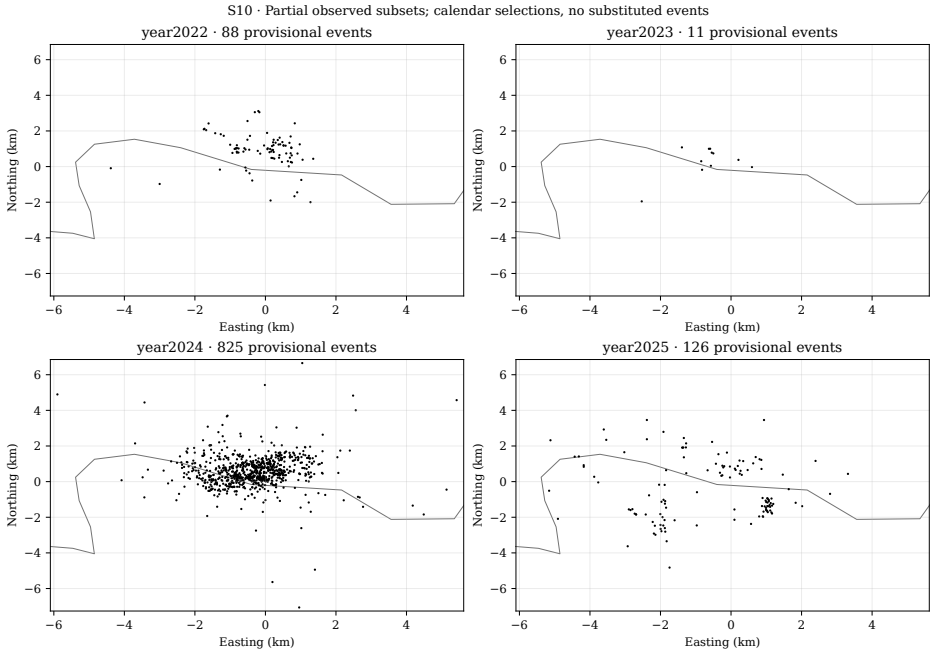
Published reference



Tan et al. (2025), tanmodelsupp.pdf, p. 21.
DOI: 10.1126/science.adw9038.

Independent result

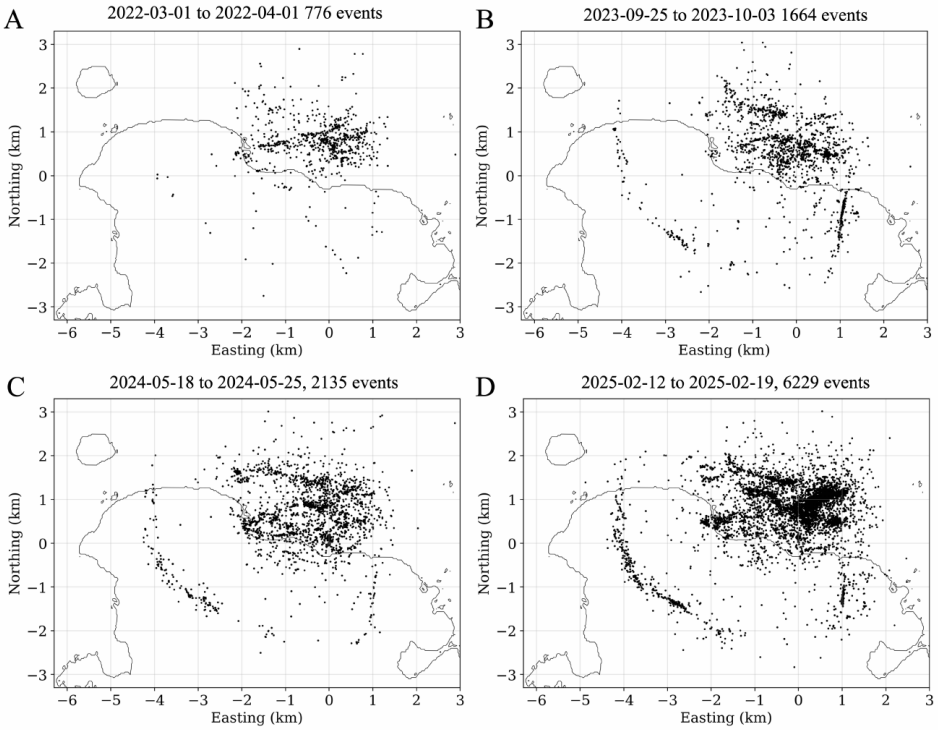
Panel S10; current provisional independent calculation — partially reproduced



Own relocated calendar-year subsets: 88 in 2022, 11 in 2023, 825 in 2024, 126 in 2025. These are sampled counts, not annual rates.

Figure S11: Four major swarms

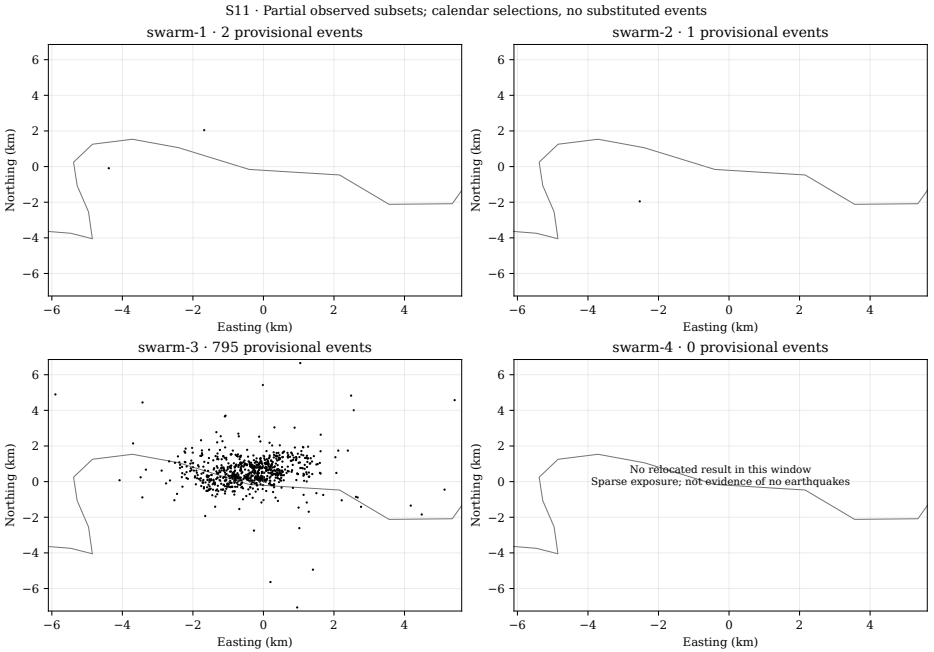
Published reference



Tan et al. (2025), tanmodelsupp.pdf, p. 22.
DOI: 10.1126/science.adw9038.

Independent result

Panel S11; current provisional independent calculation — partially reproduced



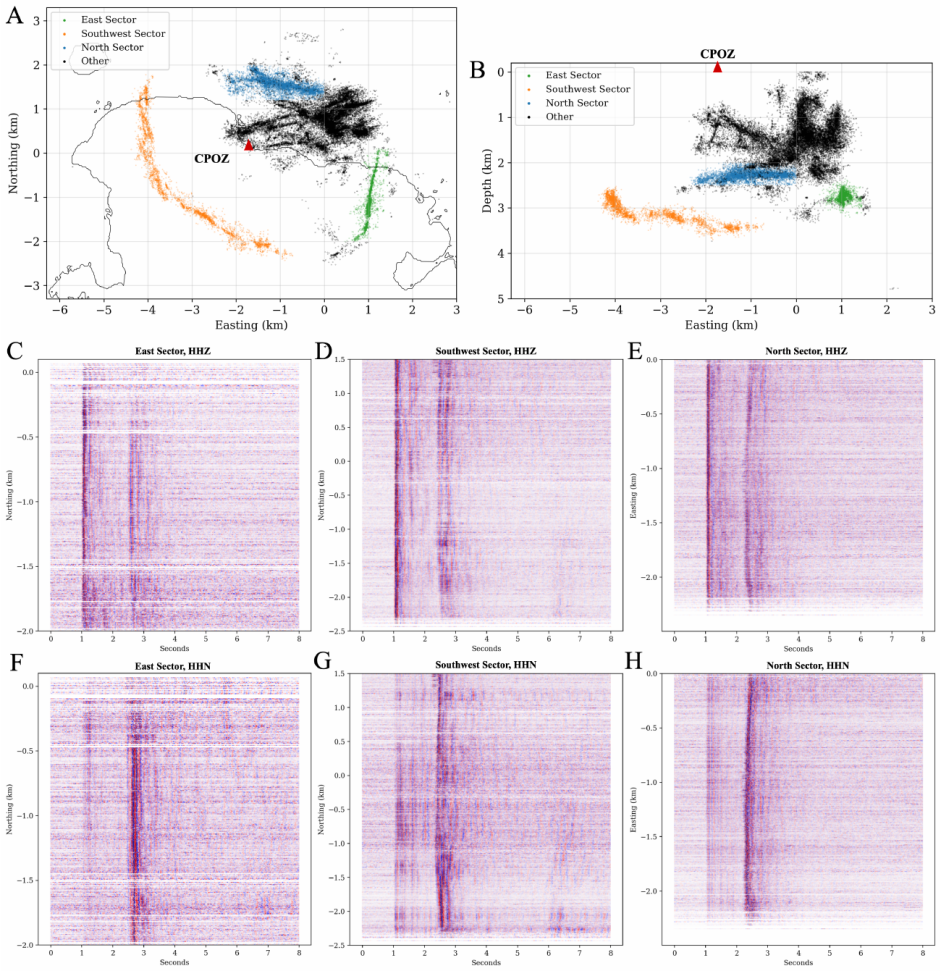
Written calendar windows with inclusive end days. Counts: swarm-1=2, swarm-2=1, swarm-3=795, swarm-4=0. Empty selections reflect available retained results, not absence of earthquakes.

Missing: S11/swarm-4 — not reproduced

S11/swarm-4: no retained independently relocated event in these written calendar windows. Check the coverage and selection ledgers; this is not evidence of no earthquakes.

Figure S12: Ring sectors and CPOZ common-receiver gathers

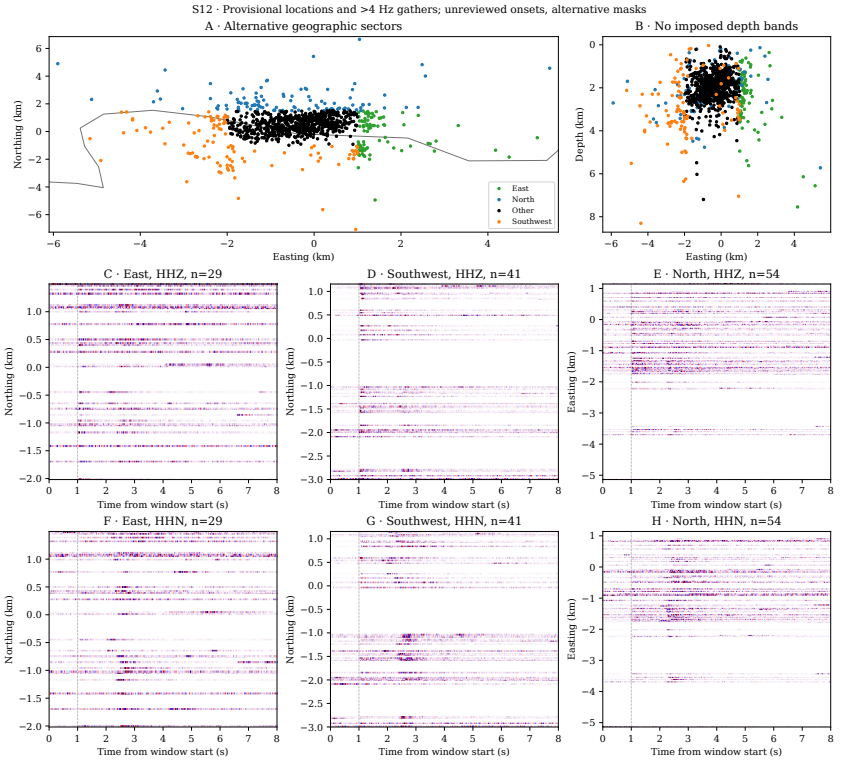
Published reference



Tan et al. (2025), tanmodelsupp.pdf, p. 23.
DOI: 10.1126/science.adw9038.

Independent result

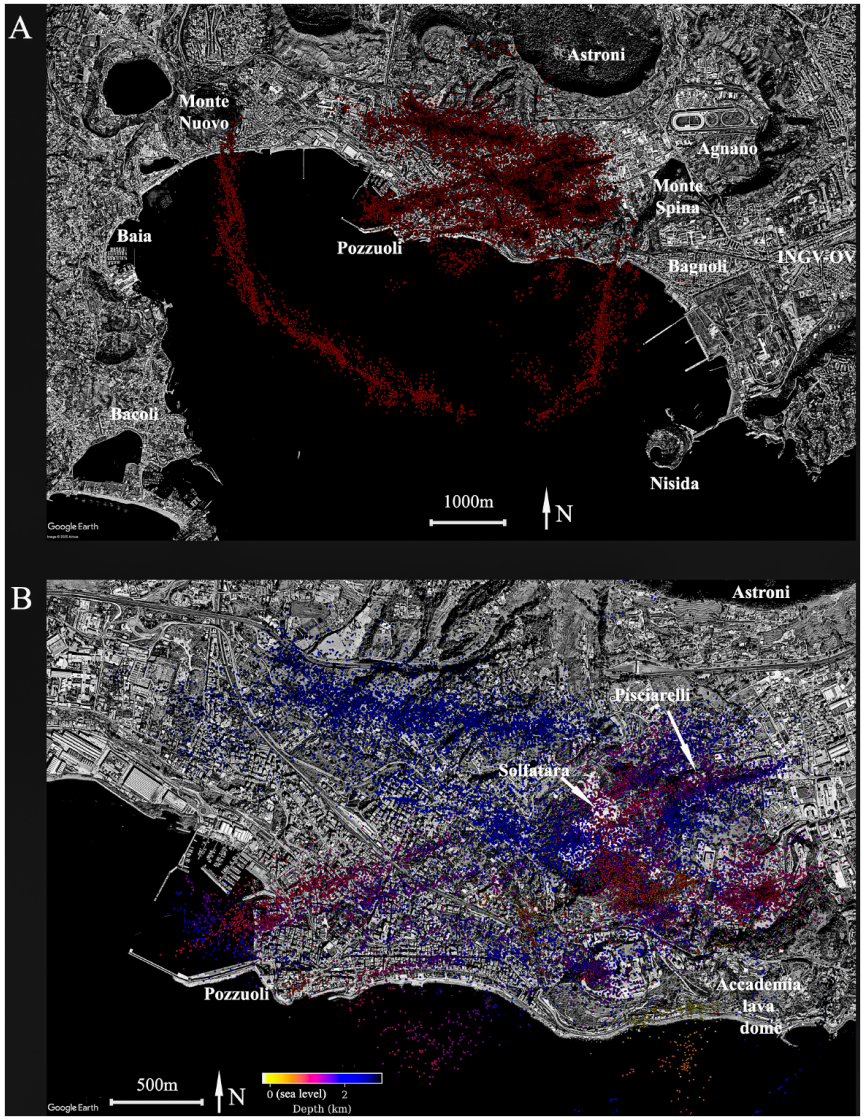
Panel S12; current provisional independent calculation — partially reproduced



1168 observed CPOZ component traces for 584 events, aligned on own P picks and filtered above 4 Hz. Explicit alternative sectors, no interpolation between traces. Onsets and absolute depths remain unvalidated.

Figure S13: Relocated seismicity on satellite imagery

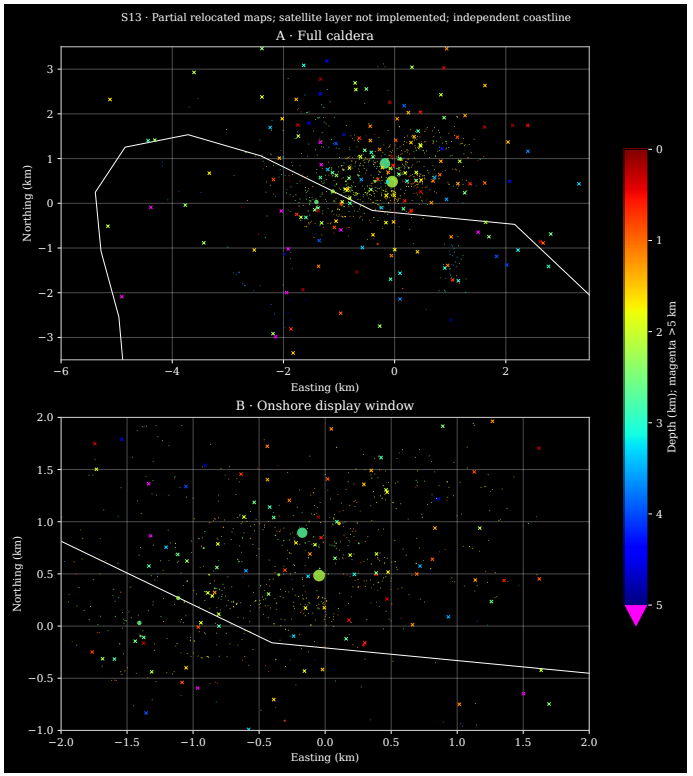
Published reference



Tan et al. (2025), tanmodelsupp.pdf, p. 24.
DOI: 10.1126/science.adw9038.

Independent result

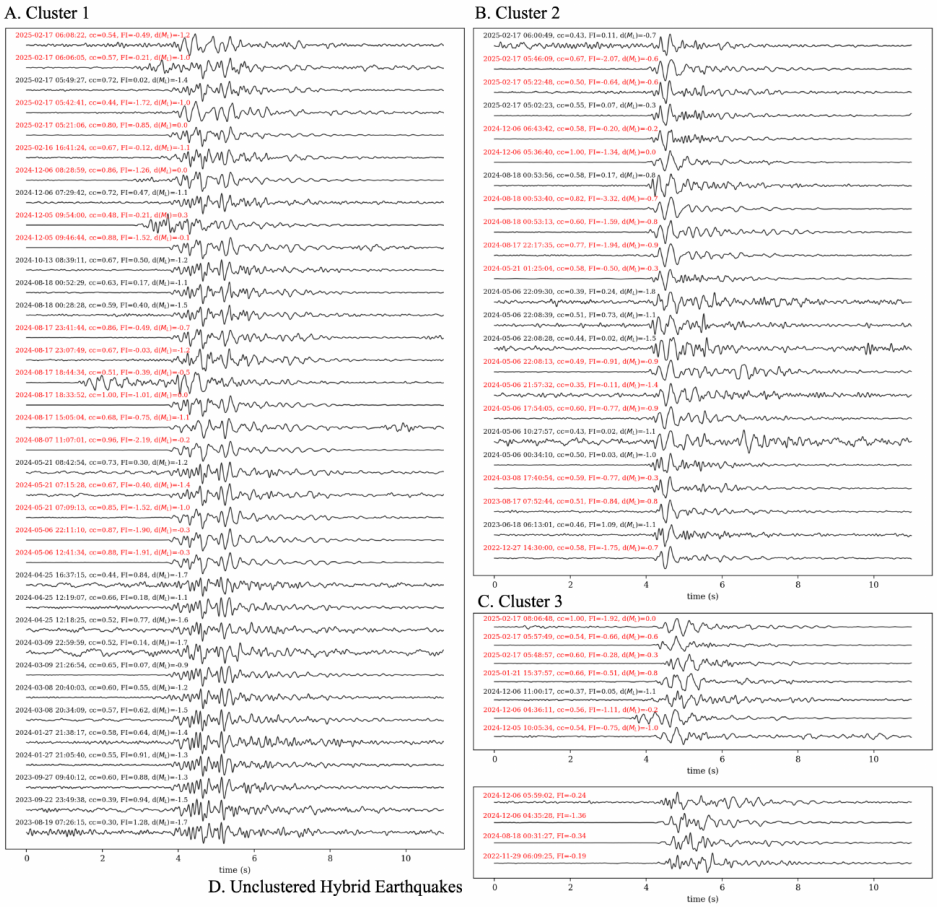
Panel S13; current provisional independent calculation — partially reproduced



Current own provisional locations on an independent coarse coastline. No satellite layer or published pixels used.

Figure S14: Hybrid-event clusters

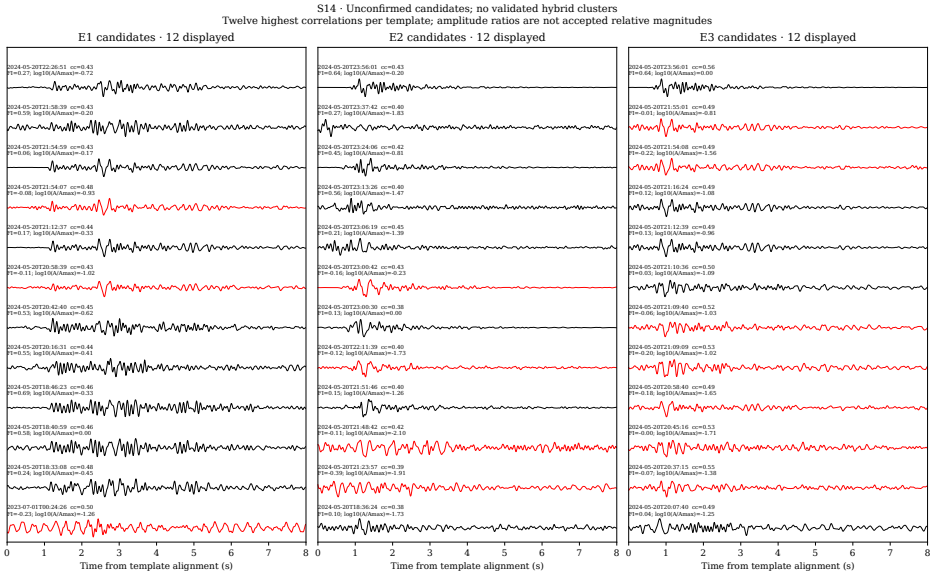
Published reference



Tan et al. (2025), tanmodelsupp.pdf, p. 25.
DOI: 10.1126/science.adw9038.

Independent result

Panel S14; current provisional independent calculation — partially reproduced



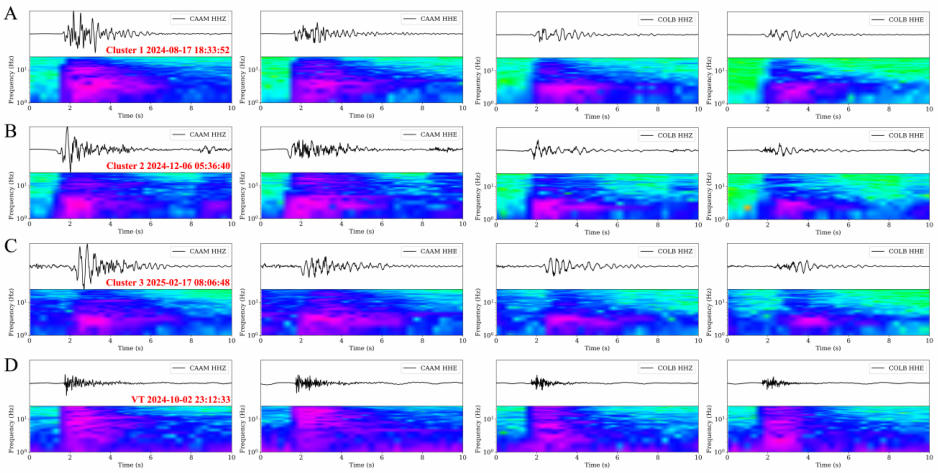
1843 unconfirmed template-score crossings over 140.656 common CAAM/COLB hours. Twelve highest scores per template displayed from original waveforms with measured frequency indices and amplitude ratios. No confirmed cluster or accepted hybrid magnitude. AI inspection of 36 displayed crossings found 4 preceding-packet tails and 3 ongoing-activity windows; the other transients remain unclassified. No human review.

Missing: S14/unclustered — not reproduced

S14/unclustered: no independent unclustered hybrid sample or validated family assignment. The three-template threshold scan cannot itself supply that class.

Figure S15: Hybrid and shallow-event waveforms

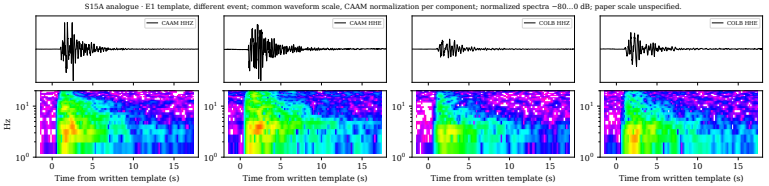
Published reference



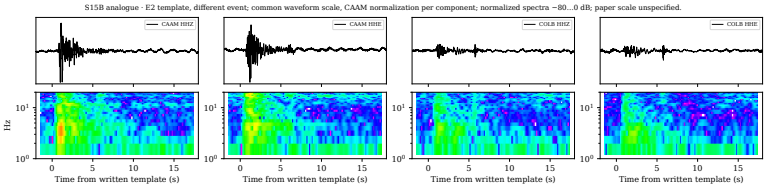
Tan et al. (2025), tanmodelsupp.pdf, p. 26.
DOI: 10.1126/science.adw9038.

Independent result

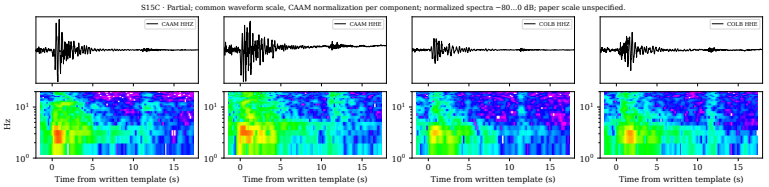
Panel S15A analogue: different documented E1 event — partially reproduced



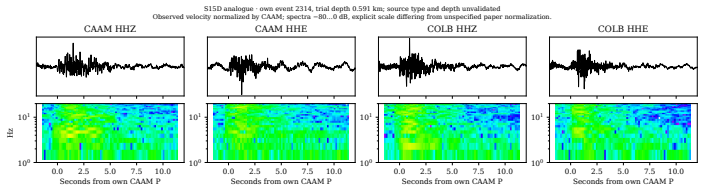
Panel S15B analogue: different documented E2 event — partially reproduced



Panel S15C: documented E3 event — partially reproduced



Panel S15D analogue: own shallow trial, source class unvalidated — partially reproduced



A/B use different written-method E1/E2 examples; C uses the documented E3 template. D: Own waveform example selected from a provisional depth below 0.6 km near CAAM. It is not the published event, and neither extremely shallow depth, VT source type, nor west-Accademia membership is validated.